

Neonatal Short Bowel Syndrome: Population-Based Estimates of Incidence and Mortality Rates

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Background: Congenital or acquired neonatal short bowel syndrome (SBS) carries significant morbidity and mortality rates. No accurate population estimates of incidence and mortality exist because of differences in definition, follow-up, and regional referral patterns.

Methods: A retrospective cohort study was performed involving 175 surgical neonates admitted to our institution from January 1, 1997 to December 31, 1999 and followed up until July 1, 2001. Institution and population-based estimates of incidence and mortality were performed using postcensal population figures (1997) from Statistics Canada.

Results: The overall incidence of SBS was 22.1 per 1,000 neonatal intensive care unit (NICU) admissions (95% CI = 15.3, 28.9) and 24.5 per 100,000 live births (95% CI = 12.1, 36.9). The incidence was much greater in premature infants (less than 37 weeks). The SBS case fatality rate was 37.5%

(95% CI = 22.5, 52.5) and the cause-specific and proportional mortality rates (for children less than 4 years old) were 2.0 of 100,000 population per year (0.4 to 3.6/100,000/year) and 1.4% (0.3% to 2.6%), respectively.

Conclusions: Patients with neonatal SBS pose a complex management challenge and are responsible for a significant cost to the health care system. To our knowledge, this study represents the first population-based estimates for neonatal SBS incidence and mortality rates. Accurate estimates will assist clinicians in counseling parents, allocating resources, and planning clinical trials.

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INDEX WORDS: Short bowel syndrome, intestinal failure, incidence, mortality, outcomes research.

SHORT BOWEL SYNDROME (SBS) is defined as the spectrum of malabsorption that occurs after resection of a major portion of the small intestine for congenital or acquired lesions.^{1,2} It is also defined as the need for prolonged parenteral nutrition secondary to intestinal failure after bowel resection.³ These patients suffer a multitude of complications secondary to long-term hospitalization and prolonged parenteral nutrition.

Accurate estimation of SBS incidence and mortality remains difficult because of variations in the SBS definition between studies, lack of comprehensive follow-up, and inability of tertiary institutions to clearly define their study population because of complex referral patterns. Estimates of incidence and mortality would be helpful in enabling health care personnel to appropriately counsel parents, allocate resources, and plan clinical trials.

The Canadian Collaborative Study Group⁴ estimated the incidence of SBS across Canada as 4.8 per million population per year, but the calculation was based on a sample size of only 11 infants. This was partly secondary to poor contribution from larger institutions with the critical mass of patients. Studies examining the need for small bowel transplant have suggested an incidence of 2 to 3 patients per million population per year, of which, half are pediatric.⁵

Mortality rates vary tremendously or are not presented at all. Because patient selection often excludes early deaths, there could be an overestimation of SBS survival in the literature. No population-based mortality estimates have ever been reported.

The purpose of this study was to determine accurate estimates of SBS incidence and mortality rates at both the institution and population level.

MATERIALS AND METHODS

A retrospective cohort study was performed including all neonates admitted to the neonatal intensive care unit (NICU) at The Hospital for Sick Children (HSC) in Toronto between January 1, 1997 and December 31, 1999 with abdominal pathology requiring laparotomy. The conclusion of the study follow-up period was July 1, 2001. The study cohort was divided into patients with SBS (n = 40) and surgical neonates who did not have SBS (n = 135). The Canadian Association of Paediatric Surgeons (CAPS) defines SBS as the need for total parenteral nutrition (TPN) greater than 42 days after bowel resection, or a residual small bowel length of less than 25% expected for gestational age.⁵ This is the definition used for this study. Definitions based solely on duration of parenteral nutrition create bias because they select for survivors and exclude patients who die early.

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Neonates with pyloric stenosis or severe life-threatening anomalies (eg, cardiac), chromosomal abnormalities/syndromes (eg, trisomy 13, 18), or comorbid conditions (eg, grade III/IV intraventricular hemorrhage) that would preclude surgical management were excluded.

Data sources included the NICU electronic patient chart and hospital records. When necessary, follow-up status was determined by communication with the patient's primary care physician or by telephone contact with the patient's legal guardian. All data abstraction was performed by a single Neonatal Nurse Practitioner. This individual was blind to the study objectives to minimize observation bias.

Continuous variables were summarized as means with SDs. The 2-sample independent *t* test was used for comparison of means, and 95% confidence intervals were included as a measure of precision. Categorical variables were summarized with percentages. An alpha value of 0.05 was considered statistically significant.

Incidence Risks and Mortality Rates

Incidence risks and mortality rates were determined at both the institution (NICU) and the population level.

Defining the Population at Risk (The Denominator)

Institution. The population at risk for neonatal SBS were newborns admitted to the NICU during the 2-year enrolment period (*n* = 1,811). The NICU at The Hospital for Sick Children is a combined medical and surgical unit, therefore, all neonatal admissions are accounted for.

NICU admissions were also stratified by gestational age (>37 weeks [term, *n* = 963] and less than 37 weeks (premature; *n* = 848)) to investigate the relationship between SBS incidence and newborn maturity.

Population. The Ministry of Health in Ontario operates a regionalized system of health care. There are 7 administrative areas across the province. Within those regions are 5 pediatric centers capable of providing speciality surgical services to infants and children. All 5 pediatric centers are located in the southern portion of the province, and each has a designated local and regional catchment area. The more sparsely populated, but vast areas of the northern province send pediatric patients to one of the 5 assigned centers or to the closest pediatric centre (which may be out of province).

The Hospital for Sick Children in Toronto offers the largest pediatric surgical service across the province. The catchment area for HSC is called the Central East Referral Region covering a population of 5,172,532 and an area of 53,501 square kilometers (Fig 1).⁶

The catchment for HSC represents 46% of the population of Ontario. Metropolitan (Metro) Toronto falls within this catchment and represents 37.9% of the provincial population (4,263,755 of 11,260,419).

The infant population is the population at risk for neonatal SBS, not the whole population that includes adults. Population figures from the period of study were obtained from postcensal data from Statistics Canada (last census was 1996). Postcensal population figures were only available for 1997; therefore, a single-year calculation was performed.

A decision was made to base population estimates on figures from Metropolitan Toronto only. The main reason was to be certain of complete capture of the population at risk. Resident newborns born within metropolitan Toronto will almost certainly be admitted to HSC. It is possible that newborns from outside metro Toronto but within the catchment could be referred to other centers leading to an underestimate of incidence. Fortunately, metro Toronto comprises more than 80% of the catchment population. Table 1 summarizes the population figures used for the incidence and mortality calculations.

Defining the Cases (The Numerator)

Institution-based estimates of incidence were based on a total of 40 SBS cases (37 premature and 3 term). Mortality estimates used the

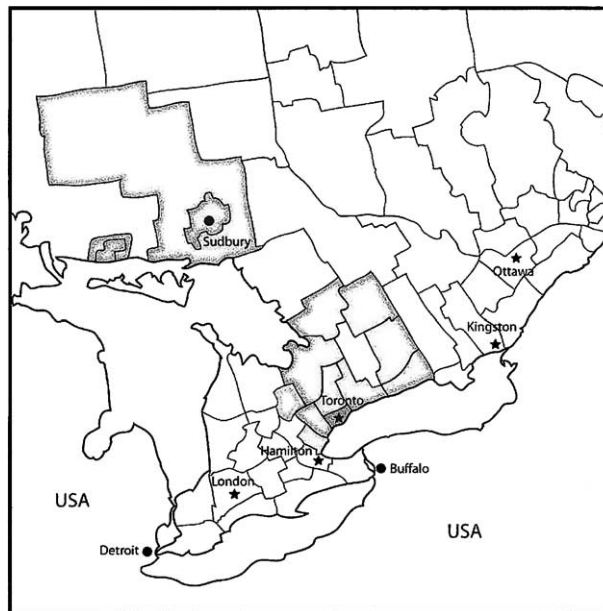


Fig 1. Catchment for the Hospital for Sick Children. Central East Referral Region indicated by shaded areas. The catchment is not geographically contiguous. Eighty percent of the catchment area is within metropolitan Toronto.

number of "all-cause" and "disease-specific" deaths for the total cohort as well as the SBS and No-SBS groups.

Population-based estimates only considered patients residing in Metropolitan Toronto (138 of 175 [78.9%]; Table 2). Because postcensal figures were only available for 1997, only the SBS cases from metro Toronto admitted in 1997 were included in the incidence calculations (total, 15; premature, 13; term, 2). There were 6 deaths in SBS patients originating in metro Toronto in 1997.

To ensure complete capture of possible cases, data from the Canadian Institute for Health Information (CIHI) were accessed. The CIHI database collects demographic data for each hospitalization, as well as clinical data (including ICD-9 diagnoses and surgical procedures) for all patients discharged from any hospital in Ontario. Submission of data to CIHI is mandatory for all hospitals in Canada. In 1997, no patient residing in metro Toronto received abdominal surgery for a condition that led to short bowel syndrome in an institution other than The Hospital for Sick Children.

Incidence Estimates

Incidence quantifies the number of new cases of disease that develop in a population of individuals at risk during a specified time interval.⁷ There are 2 types of incidence measures, cumulative incidence and incidence density. *Cumulative incidence* is the proportion of people in whom a disease develops over a specified period of time.⁸ It assumes the whole population at risk has been followed up for the specified time interval. Often, study participants are enrolled over a period of time, and the time available to observe an outcome is variable between subjects. *Incidence density* accounts for these situations in which people are susceptible for different durations. The denominator for incidence density is not persons-at-risk for a specific time period but person-time at risk of the event.⁸

Neonatal SBS is secondary to congenital or acquired conditions that put the individual at risk immediately after birth or very shortly thereafter. By definition, for patients who have more than 25% of residual small bowel after surgery, SBS is diagnosed after 42 days of

Table 1. Population, Birth, and Death Figures from 1997 Postcensal Data

	Population	Total LB	Popn 1-4 yr	<37 wk LB	>37 wk LB	Infant Deaths	1-4 yr Deaths	0-4 yr Deaths*
Metro	4,263,755	61,249	238,611	3,675	57,574	343	71	414
Ontario	11,260,419	133,004	614,655	9,310	123,694	728	153	881
Canada	30,003,955	348,598	1,583,720	24,402	324,196	1,928	455	2383

Abbreviations: LB, live births; <37 wks, premature; >37 wks, term; Popn, population; Population 1-4 yrs, total population (0-4 yr) minus live births; Infant deaths, death of live born baby before the 364th completed day of life.

*Deaths 0-4 yrs, Infant deaths plus deaths 1-4 yrs (calculated).

parenteral nutrition. Therefore, a long period of follow-up is not required to capture the diagnosis. For these reasons, cumulative incidence was believed to be the more appropriate measure.

Institution estimates are based on the 2-year enrolment period. The overall incidence, as well as the incidence in both premature and term NICU admissions was calculated. Estimates are presented per 1,000 NICU admissions.

Population-based estimates are based on the number of live births in metro Toronto. As mentioned, postcensal birth figures were only available for 1997; therefore, the number of SBS patients enrolled during 1997 and residing in metropolitan Toronto were included in a single-year calculation. The overall, premature, and term incidence per 100,000 live births is presented. Ninety-five percent confidence intervals were included with each estimate.

Mortality Estimates

Mortality rate expresses the incidence of death in a particular population during a specified period of time.⁸ Four measures of mortality were performed.

Institution or study-based estimates included the case fatality rate and the mortality rate. The case fatality rate is the number of deaths from a disease divided by all cases of that disease. It represents the proportion of fatal cases among affected individuals. The case fatality rate is a measure of disease severity.⁷

The length of follow-up was different for all study participants; therefore, the time available to observe a subject's death was variable. As a result, to calculate the study mortality rate, the denominator was expressed in person-time units. Person-years follow-up was calculated for the whole cohort (381.5 person-years), the SBS group (64.4 person-years), and the No-SBS group (317.1 person-years) separately. Time zero was defined as the date of primary surgery, and the length of follow-up was determined by the date of death, date of study closure, or last date seen for those lost to follow-up.

"All-cause" and "disease-specific" mortality per 100 person-years were determined separately for the total cohort, the SBS group, and the No-SBS group. Ninety-five percent confidence intervals were included for case fatality rate and study mortality rate. Disease-specific mortality was defined as death from hepatic failure, sepsis, or multiple organ dysfunction (MOD).

Population-based estimates included the cause-specific mortality rate and the proportional mortality rate. Patients with SBS may not die during infancy (<1 year old); therefore, the population at risk was defined as children less than 4 years of age from metro Toronto in 1997 (as with the incidence estimates, accurate postcensal death figures were only available for 1997). The cause-specific mortality rate estimates the

number of deaths from a specific cause among all individuals in a population during a given time period.⁷ It has been presented per 100,000 population (<4 years of age) with the 95% confidence interval. The proportional mortality rate expresses the number of deaths from a specific cause divided by the total number of deaths in a population during a given time period.⁹ It is presented as a percentage with 95% confidence interval.

Ethical Approval

Ethical approval was granted by the Institution Review Board at The Hospital for Sick Children after submission of the study protocol.

RESULTS

Patient Population

During the 2-year enrolment period, 1,811 patients were admitted to the NICU. Two hundred nineteen patients had abdominal operations, but 44 were excluded for reasons outlined above. One hundred seventy-five patients made up the study cohort. The cohort represents 9.6% (175 of 1,811) of all NICU admissions. Follow-up continued to the date of death or study closure (July 1, 2001) in all but 2 patients. Both patients were out-of-province referrals and were transferred to an institution closer to home postoperatively. These patients were censored at the time of discharge.

Table 3 summarizes the study population. The SBS patients were significantly more premature than the No-SBS patients (30.7 weeks v 35.9 weeks). As a result of their prematurity, the SBS group also had a significantly lower birth weight. The most common etiologies of SBS were necrotizing enterocolitis, intestinal atresia, abdominal wall defects, volvulus, and complicated meconium ileus.

Incidence Estimates

The overall institutional incidence of SBS is estimated at 22.1 per 1,000 admissions, and the true value lies between 15 and 29 per 1,000 admissions with 95% confidence (Table 4). Because our NICU admits approximately 800 to 1,000 babies annually, this value also approximates the number of SBS cases admitted per year. When SBS cases and NICU admissions are stratified by gestational age, a much larger incidence of SBS is noted in premature newborns (43.6 of 1,000 v 3.1 of 1,000 admissions). Premature infants are at a much higher risk of necrotizing enterocolitis, which is a lead-

Table 2. Regional Origin of Study Cohort

	SBS	No-SBS	Total
Metro Toronto	29	109	138 (78.9)
Remainder of catchment	8	20	28 (16.0)
Outside catchment referrals	3	6	9 (5.1)

NOTE. Values are expressed as numbers with (percentages).

Table 3. Cohort Characteristics

	SBS (n = 40)	No-SBS (n = 135)	Total (n = 175)	P Value	95% CI
GA (wk)	30.7 (4.6)	35.9 (4.8)	34.7 (5.2)	<0.0005	(-6.9, -3.5)
BWt (g)	1660 (939)	2580 (993)	2370 (1052)	<0.0005	(-1269, -572)
Males (%)	22 (55.0)	68 (50.4)	90 (51.4)		
Etiology					
NEC	14 (35)	27 (20.3)			
Atresia	4 (10)	28 (21.1)			
AWD	5 (12.5)	17 (12.8)			
Volvulus	4 (10)	10 (7.5)			
Hirschsprung's	1 (2.5)	26 (19.5)			
Meconium ileus	8 (20)	11 (8.3)			
Other	4 (10)	14 (10.4)			
Follow-up (d)	588 (428)	857 (463)	796 (468)		

NOTE. Values are expressed as mean (SD) or number with (%); 95%CI = difference in means.

Abbreviations: GA, gestational age; BWt, birth weight; AWD, abdominal wall defects.

ing cause of neonatal SBS. Babies with congenital anomalies are also more likely to be delivered prematurely.

The population-based estimates show a similar pattern. The overall incidence of SBS is 24.5 per 100,000 live births. The incidence is much greater in premature live births.

The 95% confidence intervals for term neonates include zero for both NICU and population values. This is secondary to the small number of term patients with SBS. Only 3 of 40 SBS cases occurred in term neonates. For the population estimates, 2 of the 3 SBS cases originated from metro Toronto in 1997.

The number of live births in Canada in 1997 was 348,598. Ignoring other regional differences, the estimated number of SBS cases nationwide would have been 83.7.

Mortality Estimates

The overall mortality rate during the study follow-up period was 33 of 175 (18.9%). Fifteen of 40 SBS patients (37.5%) and 18 of 135 No-SBS patients (13.3%) died. Cause of death is outlined in Table 5. Hepatic failure was responsible for 9 of 15 (60.0%) SBS deaths and none of the No-SBS deaths. "Disease-specific" mechanisms (hepatic failure, sepsis, multiorgan dysfunction) caused 13 of 15 (86.7%) SBS deaths and 12 of 18 (66.6%) No-SBS deaths.

Table 6 summarizes the mortality estimates. The case fatality rate for SBS was 15 of 40 (37.5%). The 95%

confidence interval suggests the true value was between 22.5% and 52.5%. The study mortality rate was calculated for the SBS group, No-SBS group, and total cohort separately. The overall disease-specific mortality rate for all surgical neonates was 6.6 per 100 person-years (95% CI = 4.1 to 9.0 per 100 person-years). The disease-specific mortality rate is 5-fold greater in SBS patients compared with No-SBS patients (20.2 per 100 person-years v 3.8 of 100 person-years).

Ten of 15 SBS deaths occurred in patients originating from metro Toronto. Six of the 10 deaths occurred during 1997. The population of children from metro Toronto between 0 and 4 years of age was 299,860 in 1997 (Table 1). Thus, the cause-specific mortality rate in 0 to 4 year olds was 2.0/10⁵/yr (95%CI = 0.4 to 3.6/10⁵/yr). Overall, there were 414 child deaths in Metro Toronto in 1997 (Table 1). The proportional mortality rate was 6 of 414 or 1.4% (95% CI = 0.3% to 2.6%). These results indicate that SBS is a small contributor to child mortality.

DISCUSSION

Study Design

A retrospective cohort study allows determination of disease incidence directly, as it investigates the relationship between exposure and disease.⁸ The major limita-

Table 4. Institution and Population-Based Incidence Estimates

	NICU* (95% CI)	Population† (95% CI)
Overall	22.1/1000 (15.3, 28.9)	24.5/10 ⁵ (12.1, 36.9)
Premature (<37 wk)	43.6/1000 (29.9, 57.3)	353.7/10 ⁵ (161.8, 545.7)
Term (>37 wk)	3.1/1000 (-0.4, 6.6)	3.5/10 ⁵ (-1.3, 8.3)

*Per 1,000 NICU admissions.

†Per 100,000 live births.

Table 5. Cause of Death

	SBS (n = 40)	No-SBS (n = 135)	Total (n = 175)
Respiratory	2 (5)	2 (1.5)	4 (2.3)
Cardiac	0	3 (2.2)	3 (1.7)
Hepatic	9 (22.5)	0	9 (5.1)
MOD	1 (2.5)	6 (4.4)	7 (4.0)
Sepsis	3 (7.5)	6 (4.4)	9 (5.1)
Other	0	1 (0.7)	1 (0.6)
Total	15 (37.5)	18 (13.3)	33 (18.9)

NOTE. Values are expressed as number with (percent).

Abbreviation: MOD, multiple organ dysfunction.

Table 6. Mortality Estimates

	SBS (95% CI)	No-SBS (95% CI)	Total (95% CI)
Study Mortality Rate*			
Disease-specific†	20.2 (10.4, 30.0)	3.8 (1.7, 5.9)	6.6 (4.1, 9.0)
All-cause	23.3 (13.0, 33.6)	5.7 (3.1, 8.2)	8.7 (5.8, 11.5)
Cause-Specific mortality rate‡			2.0 (0.4, 3.6)
Proportional mortality rates§			1.4% (0.3, 2.6)
Case fatality rate			37.5% (22.5, 52.5)

*Study mortality rate expressed per 100 person-years.

†Disease-specific mortality = death from hepatic failure, sepsis, multiple organ dysfunction.

‡Cause-specific mortality rate expressed per 100,000 population (0-4 yrs old).

§Proportional mortality rate expressed as % of deaths in 0-4 yr olds (with 95% CI).

||Case fatality rate expressed as % with (95% CI).

tion, however, of a retrospective cohort study is bias. Bias represents a systematic error that results in incorrect estimation of the association between exposure and disease.⁷ Every type of analytic study is susceptible to bias. A reader must decide if bias distorts the results sufficiently to disregard the conclusions.

Selection bias is an error in the process of identifying the study population.¹⁰ It is a particular problem in retrospective cohort studies because the exposure and outcome have already occurred at the time of participant enrolment. The advantage of this study was that the whole population of surgical neonates was enrolled—not a sample.

Surgical research is particularly vulnerable to selection bias because the indications for surgery can be subjective. Even when guidelines for operation are clear (as in interventional trials), an individual surgeon may or may not decide surgery is indicated. Thus, a study could be criticized because the surgical cohort may not represent the true population. The difference with this study is that essentially all congenital anomalies of the neonatal gastrointestinal tract need surgical intervention. Hence, the indications are clear. Indications for surgery in babies with NEC, however, are more subjective. Therefore, a neonate's enrolment in the cohort depends on whether the surgeon decided a laparotomy was necessary. This clinical decision directly affects the number of cases. Typically, neonates with known abdominal pathology do not receive surgical therapy only if their comorbid disease makes surgery unfeasible. This potential selection bias needs to be considered in all surgical research and interpreted to determine if the results are still valid.

Observation bias relates to differences in the way exposure or outcome data are obtained from study groups.⁷ Measures taken in this project to minimize observation bias included the use of a single data abstractor who was blinded to the study groups and research objectives. Great effort was made to have complete follow-up. Recall bias was not an issue as the data source was based primarily on hospital records. It was also unlikely that misclassification was significant be-

cause most variables were objective findings or dates. No “proxy” variables were intentionally used from records.

Incidence Risks and Mortality Rate Estimates

To date, there has not been publication of accurate incidence and mortality estimates for SBS. In part this is secondary to the rare nature of the condition. More important, however, is the difficulty major referral centers experience in precisely defining their catchment area.

The overall institution-based incidence of SBS was 22.1 per 1,000 (15.3, 28.9) NICU admissions, whereas the population estimate was 24.5 per 100,000 (12.1, 36.9) live births. These calculations were based on a modest sample size, but they do make a useful contribution for several reasons.

The attempt to clearly define the population at risk improves the reliability of the estimates. Newborns and infants, not adults, are at risk of neonatal SBS. Incidence risks based on the whole population, like the CAPS (1999) estimate of 4.8 per 1 million population per year, can be misleading.

The decision to restrict the catchment calculations to metropolitan Toronto minimized any potential loss of patients to other centers and still provided a large population to be considered. This combined with the fact that only admissions originating from metro Toronto were included in the population calculations increases one's confidence in the quality of the results.

The risk of SBS is not constant across all age groups; therefore, incidence risks stratified by gestational age (> or <37 weeks) are helpful. Both the institution and population estimates showed the significantly higher frequency of SBS in premature babies. Institutions could use figures such as these to adjust their incidence predictions by their number of premature admissions, permitting more accurate estimates. There were few term babies with SBS in this study. The 95% confidence intervals for term babies includes zero. Repeating the estimates with a larger cohort (once more Canadian postcensal data becomes available) would be useful.

From a research perspective, knowledge of SBS incidence helps determine the feasibility of clinical trials. For example, assume investigators want to study the impact on mortality of a novel treatment versus conventional therapy. The case fatality rate for SBS in this study was 37.5% (95% CI = 22.5%, 52.5%). If investigators want to detect a 20% difference in mortality at a power of 90% with an alpha value of 0.05, they would need to recruit 833 patients into each group.⁹ This would be the minimum number of participants because it does not consider any control of confounding through stratification. The Canadian incidence of SBS is approximately 85 cases annually. Therefore, the study would not be feasible because of the long recruitment time. A multinational trial would need to be conducted.

Until now, the only mortality estimates published have been the equivalent of the case fatality rate. For mortality estimates to be accurate in observational studies an adequate follow-up period must have occurred to allow enough time for the event to happen. Also, all participants need to be followed up to prevent bias.⁸ In this study, 2 patients were lost to follow-up. Also, the study design permitted a minimum of 1.5 years of follow-up. Most of the children who died did so well within this minimum follow-up period; however, it is possible more

deaths would have occurred if the follow-up period were extended. If this is true, the mortality rates have been underestimated.

The inclusion of mortality rates per 100 person-years accounted for the variation in follow-up between participants. All available information is utilized in the calculations. This can be misleading, however, because a large number of patients followed up for a short period contributes as much to the denominator as fewer patients followed up for a long time.⁸ Calculation of mortality based on person-years at risk makes the assumption that the risk of death from SBS is constant over time.⁷ This is unlikely as the risk probably varies with patient age, the point of time in the postoperative period, and the year of study. A conscious decision was made to only investigate patients from a recent 2-year interval (1997 to 1999), to minimize the variability in medical care.

Patients with neonatal SBS pose a complex management challenge and are responsible for a significant cost to the health care system. This study attempts to present more precise population-based estimates for neonatal SBS incidence and mortality. Accurate estimates will assist clinicians in counseling parents, allocating resources, and planning clinical trials.

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